



Finite-difference Cartesian cut-cell method for hyperbolic systems

Nek Sharan*, Peter T. Brady[†] and Daniel Livescu.[‡]
Los Alamos National Laboratory, Los Alamos, NM, 87544

Finite-difference Cartesian cut-cell schemes provide a dimensionally-split discretization for fluid-flow calculations over complex geometries of engineering interest. Non-dissipative central finite-difference schemes are commonly used for high-fidelity turbulent flow simulations in canonical configurations, but their application to practical computational domains requires careful boundary treatment to ensure long-time stability, high-order accuracy, and conservation. A framework to construct cut-cell boundary closures for central finite-difference schemes is developed that ensures time stability and fixes the small-cell problem. The framework is based on the idea of deriving boundary stencils that yield an energy bound (for problems with bounded energy) and that have finite cell sizes even when the grid spacing at cut-cell boundary becomes negligibly small. Following that idea, cut-cell first- and second-derivative approximations are derived to solve hyperbolic, parabolic, and incompletely parabolic problems. The resulting schemes do not require geometry/solution reconstructions over a volume or surface to account for the embedded object's intersection with Cartesian grid cells. The stability and accuracy of the proposed method is tested for various linear and non-linear problems.

I. Introduction

Cartesian cut-cell approaches are a class of immersed boundary methods (IBMs) that allow sharp representation of solid boundaries in a fluid domain [1, 2]. A Cartesian discretization for the entire computational domain highly simplifies grid generation for fluid-flow simulations over complicated geometries and improves computational efficiency by providing a structured framework. Another class of IBMs that allow sharp solid boundaries are ghost-cell methods [3, 4] which differ from the cut-cell approaches in how the fluxes (in finite-volume schemes) or derivative approximations (in finite-difference schemes) are evaluated near an immersed boundary (IB).

Ghost-cell methods use ghost cells/points that extend inside IB, where the solution is extrapolated to account for the IB. They are typically used with a finite-difference scheme, where the ghost points allow use of uniform-grid interior stencils at boundary that circumvents the small-cell stability problem [5]. In contrast, the cut-cell methods apply biased reconstructions/stencils to determine fluxes/derivative-approximations near IB using only the fluid-cell data. As a result, the cut-cell approaches need to account for non-uniform cells near IB which can become arbitrarily small causing the small-cell problem. Cut-cell approaches are typically used with finite-volume schemes that ensure discrete conservation, an attribute often overlooked in ghost-cell approaches. The small-cell problem in finite-volume cut-cell methods is addressed by cell mixing [6], cell merging [7], cell linking [8] or flux redistribution [9] strategies that lower the order-of-accuracy of the numerical scheme locally and introduce additional complexity of index changes and bookkeeping. The present work develops a finite-difference cut-cell approach that, by construction, addresses the small-cell problem without compromising on global high-order accuracy. Moreover, the dimensionally-split calculations in a finite-difference scheme simplify implementation in higher dimensions.

*CCS-2, Los Alamos National Laboratory. AIAA Member.

[†]CCS-2, Los Alamos National Laboratory. AIAA Member.

[‡]CCS-2, Los Alamos National Laboratory. AIAA Associate Fellow.

Existing sharp-interface immersed boundary approaches are widely used for aerodynamic calculations [10, 11]. However, their application to high-fidelity turbulent flow simulations is limited by a lack of high-order accurate and non-dissipative IBM. Low-order boundary treatments or additional stabilization measures, such as artificial dissipation or filters, that compromise the global accuracy and conservation properties of the numerical scheme are often required with existing IBMs to address the numerical instabilities arising at IBs when non-dissipative centered scheme is used in the interior. Among the few existing high-order IBMs are a fifth-order accurate finite-difference ghost-cell method [12] for inviscid problems that uses an inverse Lax-Wendroff boundary treatment at inflow IBs and a third-order accurate finite-volume cut-cell method [13] for viscous problems that uses a cell clustering technique with piecewise polynomial approximation of the flow solution at IBs. The extension of the former approach to viscous problems is non-trivial and both approaches employ essentially non-oscillatory interior scheme.

High-order centered interior schemes are preferred for high-fidelity turbulent flow [14–17] and aeroacoustics [18–21] simulations because of their non-dissipative properties. A recent study [22] employed a non-linear optimization based approach [23] to derive numerically stable high-order cut-cell boundary stencils for centered interior schemes. This study examines an analytical framework to obtain provably time-stable cut-cell finite-difference schemes with centered interior schemes, following the methodology of [24, 25] for strong boundary conditions (BCs). Numerical time-stability analyses of finite-difference schemes using weak BC implementations are widely available (*e.g.* [26–28]), however similar analyses with strong BCs are rare.

The paper is organized as follows. Section II describes the key steps in the proposed finite-difference cut-cell scheme. Section III discusses the derivation of cut-cell boundary stencils to address the small-cell problem and to ensure time stability. The construction of first-derivative approximation is described in section III.A, and second-derivative in section III.B. Numerical results from the application of derived cut-cell boundary stencils are discussed in section IV, and the conclusions are provided in section V.

II. The finite-difference cut-cell discretization

Most existing cut-cell approaches use a finite-volume discretization with fluxes approximated using an upwind or essentially non-oscillatory reconstructions (*e.g.* [13, 29]). These reconstructions help stabilize simulations, especially at cut-cell boundaries, however, they introduce numerical dissipation that may unfavorably influence turbulence/mixing statistics [30–32]. To alleviate the issue, this study derives cut-cell boundary treatments for centered schemes that are numerically non-dissipative.

The key steps in the proposed finite-difference cut-cell discretization are shown schematically in figure 1. The fluid and the solid domain, denoted by Ω_f and Ω_s , respectively, intersect at the fluid-solid (cut-cell) boundary, labeled Γ in image (1). The pre-processing step involves identifying the fluid nodes, where the fluid-flow equations must be solved. The grid points shown as gray nodes in image (2) are, therefore, excluded from fluid calculations since they lie in Ω_s . To account for the fluid-solid boundary, cut-boundary nodes, colored blue in image (3), are introduced at the intersection of the solid body with Cartesian grid lines. A finite-difference derivative calculation uses one-dimensional stencils in each Cartesian direction to yield a dimensionally-split discretization. The introduction of cut-boundary nodes, however, results in non-uniform grid spacing near the fluid-solid boundary, as shown, for example, by the x - and y -direction grid spacings in the red and green dashed boxes, respectively, in image (3). Time-stable derivative approximations for such cut-cell boundaries are discussed in the next section.

Boundary conditions at cut-cell boundaries are applied in this study based on the characteristic wave directions of the hyperbolic equations. If the Dirichlet boundary data of the incoming characteristic waves are known at the cut-cell boundary, the governing equations can be solved at the fluid nodes (colored in black in all images of figure 1) using the derivative approximations discussed in the next section. However, in case of Neumann or Robin boundary data

or when the governing equations need to be solved at the cut-boundary nodes, Cartesian derivatives may be required at the boundary nodes. To discuss the derivative calculation at boundary nodes, it is useful to categorize them based on the Cartesian grid line they intersect. In image (4) of figure 1, the boundary nodes intersecting the x -grid line ($y = \text{constant}$) are colored red, whereas the boundary nodes intersecting the y -grid line ($x = \text{constant}$) are colored green. At red nodes, the x -derivative, $\partial(\bullet)/\partial x$, can be approximated from the existing fluid and boundary nodes (using the biased cut-cell boundary stencils presented in the next section) but y -derivative, $\partial(\bullet)/\partial y$, approximation requires interpolation (*e.g.* along the red lines in image (4) for respective boundary nodes). Similarly, at green boundary nodes, y -derivative can be approximated from the existing fluid and boundary nodes, whereas interpolation (along the green lines, for example) is required to calculate x -derivative.

III. Derivative stencils for cut-cell boundaries

To solve the fluid-flow governing equations near cut-cell boundary, derivative approximations that account for non-uniform grid spacing at boundary node (as observed in the red and green dashed boxes of image (3) in figure 1) are required. This section discusses derivation of cut-cell boundary stencils to ensure discrete conservation and time stability, and address the small problem.

For a well-posed initial boundary value problem of a system of partial differential equations, a discretization of spatial derivatives using a finite-difference scheme yields, via a method of lines approach [33, 34], a system of ordinary differential equations (or a semi-discretization)

$$\frac{d\mathbf{u}}{dt} = \mathcal{M}\mathbf{u} + \mathbf{b}, \quad (1)$$

with initial data $\mathbf{u}(0) = \mathbf{f}$. In (1), $\mathbf{u}(t)$ is a vector of unknowns, $\mathcal{M}$ is bounded linear operator (or matrix) referred to here as the system matrix, and $\mathbf{b}(t)$ is a vector that includes the boundary data and forcing functions. Using Definition 2.13 of [35], time-stability is defined as:

Definition 1. *The approximation (1) is time-stable if for $\mathbf{b} = 0$, there is a unique solution $\mathbf{u}(t)$ satisfying*

$$\|\mathbf{u}\|_H \leq K \|\mathbf{f}\|_H, \quad \text{or} \quad \frac{d}{dt} \|\mathbf{u}\|_H^2 \leq 0, \quad (2)$$

where K is independent of Δx , $\mathbf{f}$ and t . H denotes a symmetric positive definite (norm) matrix.

Substituting $\mathbf{b} = \mathbf{0}$ in (1) yields the time-stability condition

$$\frac{d}{dt} \|\mathbf{u}\|_H^2 = \mathbf{u}^T H \left(\frac{d\mathbf{u}}{dt} \right) + \left(\frac{d\mathbf{u}}{dt} \right)^T H \mathbf{u} = \mathbf{u}^T H (\mathcal{M}\mathbf{u}) + (\mathcal{M}\mathbf{u})^T H \mathbf{u} = \mathbf{u}^T [H\mathcal{M} + (H\mathcal{M})^T] \mathbf{u} \leq 0. \quad (3)$$

It can be shown that (3) is satisfied if either of the following sufficient conditions hold [36–38]:

- 1) There exists a norm matrix H such that $\mathbf{x}^T H \mathcal{M} \mathbf{x} \leq 0$ for all $\mathbf{x}$, or that the symmetric matrix $H\mathcal{M} + (H\mathcal{M})^T$ is negative semidefinite.
- 2) All eigenvalues of $\mathcal{M}$ have nonpositive real parts, and every eigenvalue with zero real part has an associated Jordan block of order one.

First- and second-derivative approximations on cut-cell grids that yield system matrices (for hyperbolic and parabolic problems with bounded energy) which satisfy either of the above time-stability conditions are discussed in sections III.A, and III.B, respectively.

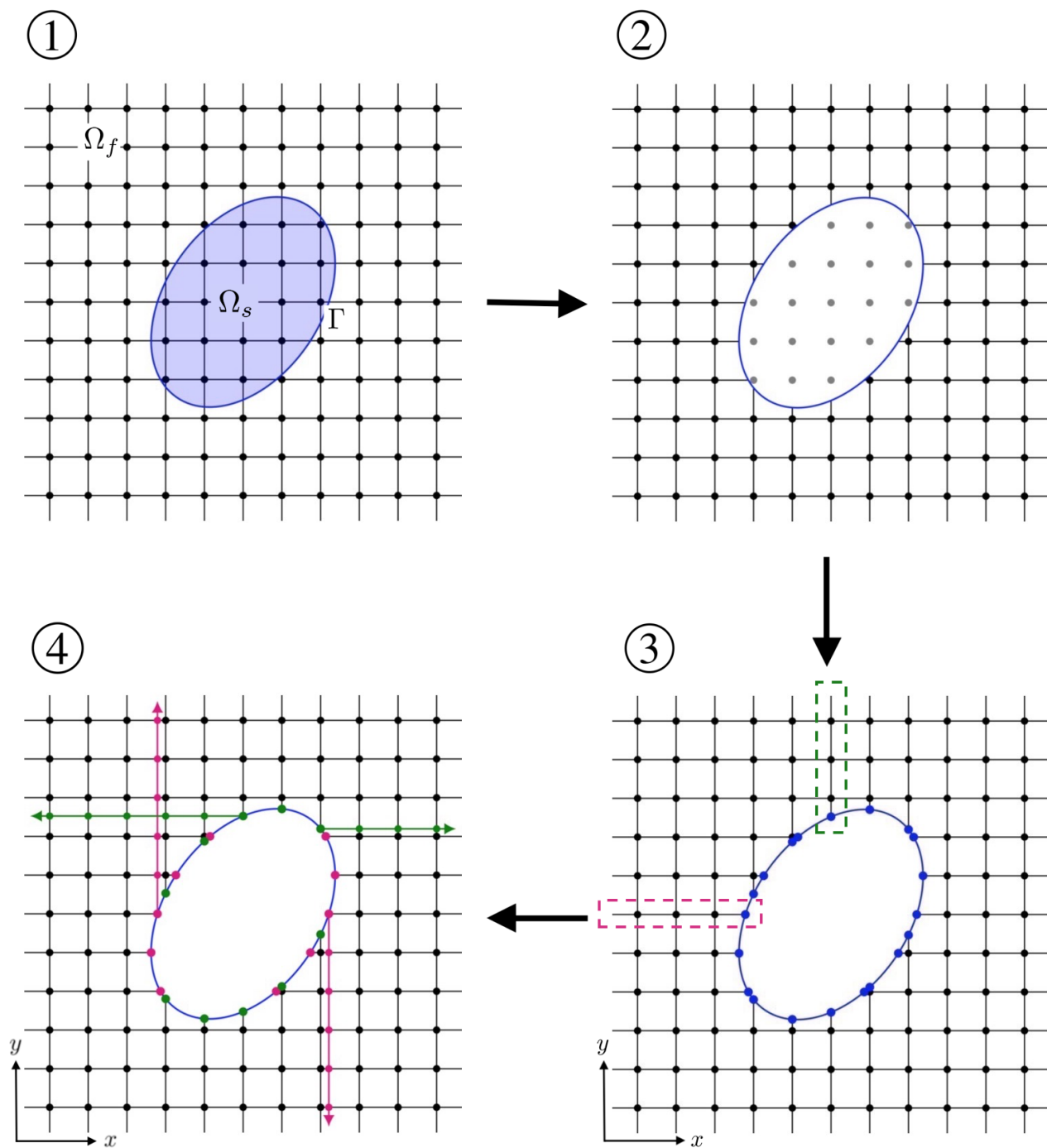


Fig. 1 Schematic of steps in the proposed finite-difference cut-cell scheme.

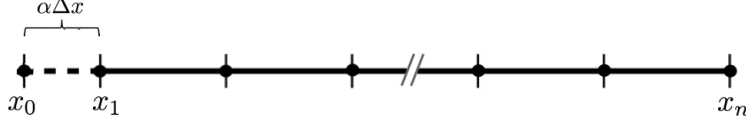


Fig. 2 One-dimensional grid with variable first grid point location, analogous to a cut-cell boundary point, $0 \leq \alpha \leq 1$.

A. First-derivative approximation

Consider the scalar hyperbolic equation

$$\frac{\partial U}{\partial t} + \frac{\partial U}{\partial x} = 0, \quad x_0 \leq x \leq x_n, \quad t \geq 0. \quad (4)$$

Let $\mathbf{u}(t) = [u_0(t) \ \cdots \ u_n(t)]^T$ denote the vector of unknowns on a one-dimensional cut-cell grid, as shown in figure 2. The first grid point location is given by

$$x_0 = x_1 - \alpha \Delta x, \quad \text{for} \quad 0 \leq \alpha \leq 1, \quad (5)$$

where Δx denotes the uniform grid spacing in the interior, *e.g.* $\Delta x = x_2 - x_1$ in figure 2. $\alpha = 1$ yields a uniform grid, whereas $\alpha = 0$ leads to coincident first and second grid points. A semi-discretization of (4) can be written as

$$\frac{d\mathbf{u}}{dt} = -D_1 \mathbf{u}, \quad (6)$$

where D_1 denotes a first-derivative approximation.

1. Discrete energy conservation

The time rate of change of energy in (4) is given by

$$\frac{d}{dt} \int_{x_0}^{x_n} U^2(x, t) dx = 2 \int_{x_0}^{x_n} U \frac{\partial U}{\partial t} dx = -2 \int_{x_0}^{x_n} U \frac{\partial U}{\partial x} dx = U^2(x_0, t) - U^2(x_n, t). \quad (7)$$

Defining a scalar product and norm as

$$(V, W) = \int_{x_0}^{x_n} V(x, t) W(x, t) dx, \quad \|V\|^2 = (V, V), \quad (8)$$

allows (7) to be rewritten as

$$\frac{d}{dt} \|U\|^2 = \left(U, \frac{\partial U}{\partial t} \right) + \left(\frac{\partial U}{\partial t}, U \right) = - \left(U, \frac{\partial U}{\partial x} \right) - \left(\frac{\partial U}{\partial x}, U \right) = U^2(x_0, t) - U^2(x_n, t). \quad (9)$$

We define a discrete scalar product and norm, analogous to (8), for real-valued vector functions $\mathbf{v} = [v_0 \ \cdots \ v_n]^T$ and $\mathbf{w} = [w_0 \ \cdots \ w_n]^T$ by

$$(\mathbf{v}, \mathbf{w})_H = \mathbf{v}^T H \mathbf{w} = \sum_{i,j=0}^{\kappa-1} h_{ij} v_i w_j \Delta x + \sum_{i=\kappa}^{n-\kappa} v_i w_i \Delta x + \sum_{i,j=n-\kappa+1}^n h_{ij} v_i w_j \Delta x, \quad (10)$$

$$\|\mathbf{v}\|_H = \sqrt{(\mathbf{v}, \mathbf{v})_H}, \quad (11)$$

where κ represents the depth of boundary stencil, and h_{ij} are the coefficients of the norm matrix H . (10) assumes a H with $\kappa \times \kappa$ boundary blocks and diagonal entries of unity in the interior. For a diagonal H , $h_{ij} = 0$ when $i \neq j$ in the interior and the boundary blocks. Using (10)-(11), a discrete analogue of (9) for semi-discretization (6) can be written as

$$\frac{d}{dt} \|\mathbf{u}\|_H^2 = \left(\mathbf{u}, \frac{d\mathbf{u}}{dt} \right)_H + \left(\frac{d\mathbf{u}}{dt}, \mathbf{u} \right)_H = -(\mathbf{u}, D_1 \mathbf{u})_H - (D_1 \mathbf{u}, \mathbf{u})_H = u_0^2 - u_n^2. \quad (12)$$

Simplifying the last equality in (12) yields

$$(\mathbf{u}, D_1 \mathbf{u})_H + (D_1 \mathbf{u}, \mathbf{u})_H = \mathbf{u}^T [HD_1 + (HD_1)^T] \mathbf{u} = u_n^2 - u_0^2. \quad (13)$$

(13) can be ensured if the matrix HD_1 satisfies the summation-by-parts property [39, 40]

$$HD_1 + (HD_1)^T = \text{diag}(-1, 0, \dots, 0, 1). \quad (14)$$

2. Discrete solution (mass/momentum) conservation

Equation (7) provides an energy conservation statement for (4), where the energy is defined as $\int_{x_0}^{x_n} U^2(x, t) dx$. A similar statement for the quantity $S = \int_{x_0}^{x_n} U dx$ is given by

$$\frac{d}{dt} \int_{x_0}^{x_n} U dx = - \int_{x_0}^{x_n} \frac{\partial U}{\partial x} dx = U(x_0, t) - U(x_n, t). \quad (15)$$

In case of fluid flow (Euler/Navier-Stokes) equations, the quantity S may represent mass or momentum with $U = \rho$ or ρu , respectively, for example. A discrete analogue of (15) for the semi-discretization (6) can be written as

$$\frac{dS}{dt} \approx \sum_{i=0}^n \left(\frac{d}{dt} H \mathbf{u} \right)_i = - \sum_{i=0}^n (HD_1 \mathbf{u})_i = u_0(t) - u_n(t), \quad (16)$$

where $(\mathbf{v})_i$ denotes the i -th component of vector $\mathbf{v} = [v_0 \ \dots \ v_n]^T$ and the entries of H constitute a quadrature for the domain $x_0 \leq x \leq x_n$. The following lemma shows that (14) ensures (16) is satisfied.

Lemma 1. *If $HD_1 + (HD_1)^T = \text{diag}(-1, 0, \dots, 0, 1)$ then $\sum_{i=0}^n (HD_1 \mathbf{u})_i = u_n - u_0$.*

Proof. The rows of a derivative approximation must sum to zero to satisfy the zeroth order accuracy constraints, i.e.,

$$\sum_{j=0}^n d_{ij} = 0, \quad 0 \leq i \leq n, \quad (17)$$

where d_{ij} denotes the i, j entry of D_1 . As a result, the sum of the entries of i -th row of $Q = HD_1$, whose i, j entry is $q_{ij} = \sum_{k=1}^n h_{ik} d_{kj}$ for $0 \leq i, j \leq n$, is

$$\sum_{j=0}^n q_{ij} = \sum_{j=0}^n \sum_{k=1}^n h_{ik} d_{kj} = \sum_{k=1}^n h_{ik} \left(\sum_{j=0}^n d_{kj} \right) = 0, \quad 0 \leq i \leq n, \quad (18)$$

where the last equality follows from (17).

$$HD_1 + (HD_1)^T = Q + Q^T = \text{diag}(-1, 0, \dots, 0, 1) \text{ implies} \\ q_{ij} = -q_{ji} \quad \text{unless} \quad i = j = 0, n, \quad (19)$$

and

$$q_{00} = -\frac{1}{2}, \quad q_{nn} = \frac{1}{2}. \quad (20)$$

Using $Q = HD_1$,

$$\sum_{i=0}^n (HD_1 \mathbf{u})_i = \sum_{i=0}^n (Q \mathbf{u})_i = \sum_{i=0}^n \sum_{j=0}^n q_{ij} u_j = \sum_{i=0}^n q_{i0} u_0 + \sum_{i=0}^n \sum_{j=1}^{n-1} q_{ij} u_j + \sum_{i=0}^n q_{in} u_n, \quad (21)$$

where, applying (19) and (18), the middle term in the RHS of (21) is

$$\sum_{i=0}^n \sum_{j=1}^{n-1} q_{ij} u_j = \sum_{i=0}^n \sum_{j=1}^{n-1} -q_{ji} u_j = - \sum_{j=1}^{n-1} \left(\sum_{i=0}^n q_{ji} \right) u_j = 0. \quad (22)$$

Applying (18) for $i = 0$ and n yields, respectively,

$$\sum_{j=0}^n q_{0j} = q_{00} + \sum_{j=1}^n q_{0j} = 0 \quad \Rightarrow \quad \sum_{j=1}^n q_{j0} = q_{00} = -\frac{1}{2}, \quad (23)$$

$$\sum_{j=0}^n q_{nj} = q_{nn} + \sum_{j=0}^{n-1} q_{nj} = 0 \quad \Rightarrow \quad \sum_{j=0}^{n-1} q_{jn} = q_{nn} = \frac{1}{2}, \quad (24)$$

where (19) and (20) are used to obtain the last equality. Substituting (22), (23) and (24) in (21) yields

$$\sum_{i=0}^n (HD_1 \mathbf{u})_i = u_n - u_0. \quad (25)$$

This completes the proof. $\square$

Note that the converse of Lemma 1 may not be true, *i.e.* $\sum_{i=0}^n (HD_1 \mathbf{u})_i = u_n - u_0$ does not guarantee $HD_1 + (HD_1)^T = \text{diag}(-1, 0, \dots, 0, 1)$.

3. Cut-cell first-derivative stencils

Consider a cut-cell configuration shown in figure 2, which arises in multi-dimensional setting as illustrated in image (3) of figure 1. The goal here is to derive first-derivative approximations, D_1 , and corresponding norm matrices, H , with following characteristics:

- 1) They must satisfy property (14) that ensures discrete energy and solution conservation.
- 2) They must not cause small-cell issues when $\alpha = 0$ in figure 2, *i.e.* when the first and second grid points coincide.
- 3) They must provide a time-stable scheme, as per Definition 1, with appropriate boundary conditions.

1 – 2 – 1 stencil

For second-order accuracy in the interior and first-order accuracy at the boundaries, a first-derivative approximation and corresponding diagonal norm matrix that satisfy (14) are

$$D_1 = \frac{1}{\Delta x} \begin{bmatrix} -\frac{1}{2\gamma} & \frac{\alpha-2\gamma+1}{2\gamma} & 1-\frac{\alpha}{2\gamma} & 0 & 0 \\ -\frac{1}{\alpha+1} & 0 & \frac{1}{\alpha+1} & 0 & 0 \\ \frac{\alpha-2\gamma}{-\alpha^2+2\alpha\gamma+2} & \frac{\alpha-2\gamma+1}{\alpha^2-2\alpha\gamma-2} & 0 & \frac{1}{-\alpha^2+2\alpha\gamma+2} & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & \ddots & \ddots & \ddots \\ & & & & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & & & -1 & 1 \end{bmatrix}, \quad (26)$$

$$H = \Delta x \begin{bmatrix} \gamma & & & & \\ & \frac{1}{2}(\alpha+1)(\alpha-2\gamma+1) & & & \\ & & -\frac{\alpha^2}{2} + \alpha\gamma + 1 & & \\ & & & 1 & \\ & & & & \ddots \\ & & & & & 1 \\ & & & & & & \frac{1}{2} \end{bmatrix}, \quad (27)$$

where γ is a free parameter (chosen as discussed below). The empty spaces in (26) and (27) denote zero entries. The above D_1 and H operators are for the cut-cell configuration shown in figure 2 that has a cut cell at the left boundary and uniform grid spacing at the right boundary, and hence the bottom-right corner of the operators use uniform-grid summation-by-parts (SBP) stencil [35, Appendix B]. In case of a cut cell at the right boundary, the boundary block at the top-left corner of (26) and (27) should be used at the bottom-right corner after “rotating” it by 180° and changing the signs in case of D_1 . “Rotating” a matrix by 180° implies the transformation

$$\begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix} \mapsto \begin{bmatrix} m_{23} & m_{22} & m_{21} & m_{20} \\ m_{13} & m_{12} & m_{11} & m_{10} \\ m_{03} & m_{02} & m_{01} & m_{00} \end{bmatrix}. \quad (28)$$

The free parameter γ is chosen to ensure positive definiteness of H , which follows if

$$0 < \gamma < \frac{1+\alpha}{2}. \quad (29)$$

The choice of $\gamma = \frac{1+\alpha}{4}$, which lies half-way in the above range, provides the following operators (in terms of α):

$$D_1 = \frac{1}{\Delta x} \begin{bmatrix} -\frac{2}{\alpha+1} & 1 & \frac{1-\alpha}{\alpha+1} & 0 & 0 & & \\ -\frac{1}{\alpha+1} & 0 & \frac{1}{\alpha+1} & 0 & 0 & & \\ \frac{1-\alpha}{\alpha^2-\alpha-4} & \frac{\alpha+1}{\alpha^2-\alpha-4} & 0 & \frac{2}{-\alpha^2+\alpha+4} & 0 & & \\ 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & & & -1 & 1 \end{bmatrix}, \quad (30)$$

$$H = \Delta x \begin{bmatrix} \frac{\alpha+1}{4} & & & & & & \\ & \frac{1}{4}(\alpha+1)^2 & & & & & \\ & & \frac{1}{4}(-\alpha^2+\alpha+4) & & & & \\ & & & 1 & & & \\ & & & & \ddots & & \\ & & & & & 1 & \\ & & & & & & \frac{1}{2} \end{bmatrix}. \quad (31)$$

Before discussing the small-cell problem (or the lack of it) with the above operators, it is worth examining their entries for upper/lower bound of α . Substituting $\alpha = 1$ in (30) and (31) provides

$$D_1 = \frac{1}{\Delta x} \begin{bmatrix} -1 & 1 & & & & \\ -\frac{1}{2} & 0 & \frac{1}{2} & & & \\ & \ddots & \ddots & \ddots & & \\ & & -\frac{1}{2} & 0 & \frac{1}{2} & \\ & & & -1 & 1 & \end{bmatrix}, \quad H = \Delta x \begin{bmatrix} \frac{1}{2} & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & \frac{1}{2} \end{bmatrix}, \quad (32)$$

which are the uniform-grid diagonal-norm SBP stencils for second-order accuracy in the interior [35, Appendix B]. And, substituting $\alpha = 0$ in (30) and (31) provides

$$D_1 = \frac{1}{\Delta x} \begin{bmatrix} -2 & 1 & 1 & & & \\ -1 & 0 & 1 & & & \\ -\frac{1}{4} & -\frac{1}{4} & 0 & \frac{1}{2} & & \\ & -\frac{1}{2} & 0 & \frac{1}{2} & & \\ & & \ddots & \ddots & \ddots & \\ & & & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & & -1 & 1 \end{bmatrix}, \quad H = \Delta x \begin{bmatrix} \frac{1}{4} & & & & \\ & \frac{1}{4} & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & & & & \frac{1}{2} \end{bmatrix}. \quad (33)$$

The first and the second grid points coincide at $\alpha = 0$. Adding the first column of D_1 to the second in (33) and adding the first two diagonal entries corresponding to the first and the second grid point in H yields the uniform-grid boundary stencils, same as (32).

Notice in (33) that, while $\alpha = 0$, none of the boundary entries in D_1 and H are disproportionately small or large, which prevents ill-conditioning of matrices, and derivatives at the (coincident) first and second grid point can be independently calculated. Cell sizes in the above diagonal-norm conservative schemes are the diagonal entries of H , which can be utilized to write a flux-based discretization with the above node-based operators. A flux-based discretization of (4) can be written as

$$\frac{du_i}{dt} = -\frac{u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}}}{h_{ii}\Delta x}, \quad 0 \leq i \leq n, \quad (34)$$

where $u_{i+\frac{1}{2}}$ are the fluxes at cell interfaces. The locations of cell interfaces for diagonal-norm operators can be determined from the diagonal entries $h_{ii}\Delta x$ of H . To give an example of transformation of a node-based discretization to flux-based discretization, the cell interface fluxes for the operators (30)-(31) are

$$u_{i+\frac{1}{2}} = \begin{cases} \frac{1}{2}u_0 + \frac{1+\alpha}{4}u_1 + \frac{1-\alpha}{4}u_2 & \text{if } i = 0, \\ \frac{1-\alpha}{4}u_0 + \frac{1+\alpha}{4}u_1 + \frac{1}{2}u_2 & \text{if } i = 1, \\ \frac{1}{2}(u_i + u_{i+1}) & \text{if } 2 \leq i \leq n-1, \end{cases} \quad (35)$$

with $u_{-\frac{1}{2}} = u_0$ and $u_{n+\frac{1}{2}} = u_n$. Since the diagonal entries of H do not approach zero as $\alpha \rightarrow 0$ in (31), the cell size remains non-zero even when $\alpha = 0$, and the small-cell problem is avoided.

2 - 4 - 2 stencil

For fourth-order accuracy in the interior and second-order accuracy at the boundaries, a first-derivative approximation and corresponding diagonal norm matrix that satisfy (14) for the cut-cell configuration shown in figure 2 are

$$D_1 = \frac{1}{\Delta x} \begin{bmatrix} d_{00} & d_{01} & d_{02} & d_{03} & d_{04} & d_{05} & d_{06} \\ d_{10} & d_{11} & d_{12} & d_{13} & d_{14} & d_{15} & d_{16} \\ d_{20} & d_{21} & d_{22} & d_{23} & d_{24} & d_{25} & d_{26} \\ d_{30} & d_{31} & d_{32} & d_{33} & d_{34} & d_{35} & d_{36} \\ d_{40} & d_{41} & d_{42} & d_{43} & d_{44} & d_{45} & d_{46} \\ & & & \frac{1}{12} & -\frac{2}{3} & 0 & \frac{2}{3} & -\frac{1}{12} \\ & & & & \ddots & \ddots & \ddots & \ddots & \ddots \\ & & & & & \frac{1}{12} & -\frac{2}{3} & 0 & \frac{2}{3} & -\frac{1}{12} \\ & & & & & & \frac{4}{49} & -\frac{32}{49} & 0 & \frac{59}{98} & 0 & -\frac{3}{98} \\ & & & & & & 0 & \frac{4}{43} & -\frac{59}{86} & 0 & \frac{59}{86} & -\frac{4}{43} \\ & & & & & & 0 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \\ & & & & & & 0 & 0 & \frac{3}{34} & \frac{4}{17} & -\frac{59}{34} & \frac{24}{17} \end{bmatrix}, \quad (36)$$

$$H = \Delta x \text{diag} \left(h_{00}, h_{11}, h_{22}, h_{33}, h_{44}, 1, \dots, 1, \frac{49}{48}, \frac{43}{48}, \frac{59}{48}, \frac{17}{48} \right), \quad (37)$$

where the cut-cell boundary parameters d_{ij} and h_{ii} , $0 \leq i \leq 4$ and $0 \leq i \leq 7$ are in Appendix A. The bottom-right corner of (36) and (37) uses the uniform-grid diagonal-norm SBP stencil for fourth-order accuracy in the interior [35, Appendix B]. The stencil in Appendix A has two free parameters, γ and k , that, like in the previous section, are chosen

to ensure positive definiteness of H , which follows if

$$\max(0, \beta_1) < \gamma < \min(\beta_2, \beta_3), \quad \text{and} \quad k \in \mathbb{R}, \quad (38)$$

where

$$\beta_1 = \frac{-59 + 72\alpha^2 + 40\alpha^3 + 6\alpha^4}{144\alpha + 120\alpha^2 + 24\alpha^3}, \quad (39)$$

$$\beta_2 = \frac{43 + 36\alpha^2 + 32\alpha^3 + 6\alpha^4}{72\alpha + 96\alpha^2 + 24\alpha^3}, \quad (40)$$

$$\beta_3 = \frac{17 + 48\alpha + 44\alpha^2 + 16\alpha^3 + 2\alpha^4}{48 + 88\alpha + 48\alpha^2 + 8\alpha^3}. \quad (41)$$

The choice of $k = 0$ and

$$\gamma = s \max(0, \beta_1) + (1 - s) \min(\beta_2, \beta_3) \quad (42)$$

with $s = \frac{147}{233}$ at $\alpha = 1$ yields the uniform-grid diagonal-norm SBP stencils for fourth-order accuracy in the interior [35, Appendix B], as desired, and these values are used for the numerical results in this paper.

4. Time-stability with the derived cut-cell derivative approximations

To solve (4) on a bounded domain, appropriate initial and boundary conditions are required, and they are given by

$$U(x, 0) = f(x), \quad U(0, t) = g(t). \quad (43)$$

It was proven in [40], for uniform grids, that a first-derivative approximation that satisfies property (14) yields a time-stable scheme for linear hyperbolic systems (with bounded energy) when the BCs are applied weakly via a simultaneous approximation term (SAT). The same can be shown for cut-cell grids using a similar approach. For example, a semi-discretization of (4) with (43) on a cut-cell grid, as shown in figure 2, with weak BC enforcement, using SAT, can be written as

$$\frac{d\mathbf{u}}{dt} = -D_1 \mathbf{u} - \tau H^{-1} \mathbf{e}_0 (u_0 - g(t)), \quad (44)$$

$$\mathbf{u}(0) = \mathbf{f}(x), \quad \mathbf{e}_0 = [1, 0, \dots, 0]^T, \quad (45)$$

where $\mathbf{u}(t) = [u_0(t) \ \dots \ u_n(t)]^T$ is the discrete solution vector. (44) has an energy estimate in the H -norm, given by

$$\frac{d\|\mathbf{u}\|_H^2}{dt} = (\mathbf{u}, \frac{d\mathbf{u}}{dt})_H + (\frac{d\mathbf{u}}{dt}, \mathbf{u})_H = -\mathbf{u}^T (HD_1 + (HD_1)^T) \mathbf{u} - 2\tau u_0^2 + 2\tau g(t)u_0 = (1 - 2\tau)u_0^2 - u_n^2 + 2\tau g(t)u_0. \quad (46)$$

If $g(t) = 0$ and $\tau \geq \frac{1}{2}$,

$$\frac{d\|\mathbf{u}\|_H^2}{dt} \leq 0, \quad (47)$$

which proves time stability (see Definition 1).

In this study, we are interested in enforcing the BCs strongly (or exactly). With strong BCs, a semi-discretization of (4) with (43) can be written as

$$\frac{d\tilde{\mathbf{u}}}{dt} = -\hat{D}_1 \mathbf{u}, \quad (48)$$

$$\mathbf{u}(0) = \mathbf{f}(x), \quad (49)$$

where $u_0(t) \equiv g(t)$ and $\tilde{\mathbf{u}}(t) = \begin{bmatrix} u_1(t) & \cdots & u_n(t) \end{bmatrix}^T$ is the solution vector without the first element, which corresponds to the grid point where the boundary data is injected. Accordingly, $\tilde{D}_1$ is the first-derivative approximation D_1 without its first row, *i.e.* $\tilde{D}_1$ is of size $n \times (n+1)$.

If we define $\tilde{H}$ as a $n \times n$ (norm) matrix obtained from H by removing its first row and column, then

$$\frac{d}{dt} \|\tilde{\mathbf{u}}\|_{\tilde{H}}^2 = -\tilde{\mathbf{u}}^T \tilde{H} \tilde{D}_1 \tilde{\mathbf{u}} - (\tilde{D}_1 \tilde{\mathbf{u}})^T \tilde{H} \tilde{\mathbf{u}}. \quad (50)$$

Assuming homogeneous boundary data ($g(t) = 0$) makes the first element of vector $\mathbf{u}$ zero, *i.e.* $u_0 = 0$. Substituting $u_0 = 0$ in (50) yields

$$\frac{d}{dt} \|\tilde{\mathbf{u}}\|_{\tilde{H}}^2 = -\tilde{\mathbf{u}}^T \tilde{H} \tilde{D}_1 \tilde{\mathbf{u}} - (\tilde{D}_1 \tilde{\mathbf{u}})^T \tilde{H} \tilde{\mathbf{u}} = -\tilde{\mathbf{u}}^T \left[\tilde{H} \tilde{D}_1 + (\tilde{H} \tilde{D}_1)^T \right] \tilde{\mathbf{u}}, \quad (51)$$

where $\tilde{D}_1$ is a square ($n \times n$) matrix obtained from D_1 by removing its first row and column. For a diagonal H matrix and D_1 if (14) is true, then

$$\tilde{H} \tilde{D}_1 + (\tilde{H} \tilde{D}_1)^T = \text{diag}(0, \dots, 0, 1). \quad (52)$$

Substituting (52) in (51) yields

$$\frac{d}{dt} \|\tilde{\mathbf{u}}\|_{\tilde{H}}^2 = -u_n^2 \leq 0, \quad (53)$$

which proves time stability of the cut-cell derivative approximations, derived in the previous section, with strong (or exact) BCs.

While the SBP property (14) ensures time-stability with diagonal-norm operators applying homogeneous BC strongly to solve a scalar advection equation, the property (14) does not guarantee an energy estimate (in terms of boundary data) for non-homogeneous BC when the BC is applied strongly. See [25] for sufficient conditions to obtain an energy estimate with non-homogeneous BC applied strongly.

B. Second-derivative approximation

We now derive the discrete approximations to $\partial^2/\partial x^2$ on a cut-cell grid. Consider the one-dimensional heat equation

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}, \quad x_0 \leq x \leq x_n, \quad t \geq 0. \quad (54)$$

The time rate of change of energy in (54) is given by

$$\frac{d}{dt} \|U\|^2 = \left(U, \frac{\partial^2 U}{\partial x^2} \right) + \left(\frac{\partial^2 U}{\partial x^2}, U \right) = - \left\| \frac{\partial U}{\partial x} \right\|^2 + U \frac{\partial U}{\partial x} \Big|_{x_0}^{x_n}. \quad (55)$$

Assuming a discrete solution vector $\mathbf{u}(t) = \begin{bmatrix} u_0(t) & \cdots & u_n(t) \end{bmatrix}^T$ on a one-dimensional cut-cell grid, as shown in figure 2, a semi-discretization of (54) can be written as

$$\frac{d\mathbf{u}}{dt} = D_2 \mathbf{u}, \quad (56)$$

where D_2 denotes a second-derivative approximation. It can be shown (e.g. [35, Section 7.2]) that a norm matrix H and a second-derivative approximation D_2 of the form

$$H D_2 = -M + E S, \quad (57)$$

where M is a positive semidefinite matrix, $E = \text{diag}(-1, 0, \dots, 0, 1)$, and S is a first derivative approximation, will satisfy a discrete analogue of (55). Further discussion on derivation of D_2 operators of the form (57) is postponed to a future publication. We here just quote the $1-2-1$ stencil for second-derivative approximation, labeled $1-2-1$ (D_2), which is used for numerical experiments in the next section.

$1-2-1$ (D_2) stencil

For second-order accuracy in the interior and first-order accuracy at the boundaries, a second-derivative approximation for the cut-cell configuration shown in figure 2 is

$$D_2 = \frac{1}{\Delta x^2} \begin{bmatrix} \frac{6}{\alpha^2+3\alpha+2} & -2 & \frac{4\alpha-2}{\alpha+1} & \frac{2-2\alpha}{\alpha+2} & 0 & & & & \\ 1 & -\frac{1}{2}\alpha(\alpha+3) & \alpha^2+2\alpha-2 & \frac{1}{2}(-\alpha^2-\alpha+2) & 0 & & & & \\ \frac{2}{\alpha^2+3\alpha+2} & 0 & -\frac{2}{\alpha+1} & \frac{2}{\alpha+2} & 0 & & & & \\ 0 & 0 & 1 & -2 & 1 & & & & \\ & & & & & \ddots & \ddots & \ddots & \\ & & & & & & 1 & -2 & 1 \\ & & & & & & 1 & -2 & 1 \end{bmatrix}, \quad (58)$$

where the empty spaces denote zero entries and uniform grid spacing is assumed at the right boundary, as in figure 2.

IV. Numerical results

In this section, numerical results from application of cut-cell boundary stencils derived in the previous section are discussed. In all cases, time integration is performed using the classical fourth-order Runge-Kutta (RK4) method. For convergence studies, the time step is taken small enough such that the temporal errors are insignificant compared to the spatial truncation errors.

A. 1-D scalar advection equation

Consider the scalar hyperbolic equation (4), on a spatial domain $x_0 \leq x \leq x_n = 1$, where $x_0 = (1-\alpha)\Delta x = x_1 - \alpha\Delta x$. The initial and boundary conditions are

$$u(x, 0) = \sin 2\pi x, \quad u(x_0, t) = g(t) = \sin 2\pi (x_0 - t). \quad (59)$$

The exact solution to the problem is $u(x, t) = \sin 2\pi (x - t)$.

Figures 3(a) and (b) show the L_∞ -error from a long-time simulation using the $1-2-1$ and $2-4-2$ stencils, respectively. A low and a high value of CFL numbers are used with various values of α to show that the derived boundary stencils do not have the small-cell problem. The error remains constant with time indicating a time-stable behavior. As expected, errors from the $2-4-2$ scheme are smaller than that from the $1-2-1$ scheme. Increasing the CFL number from 0.3 to 0.8 does not have a visible influence on error profile indicating that the spatial truncation error dominates in these runs.

Figures 4(a) and (b) show the L_∞ -error and the convergence rates of the two schemes for various values of α . As expected, the $1-2-1$ scheme converges with second-order accuracy and the $2-4-2$ scheme converges with third-order accuracy [41]. The error profiles in figures 3 and 4 suggest that the prefactor in leading-order truncation error term does not vary much with α and, therefore, the error magnitude is similar for various values of α .

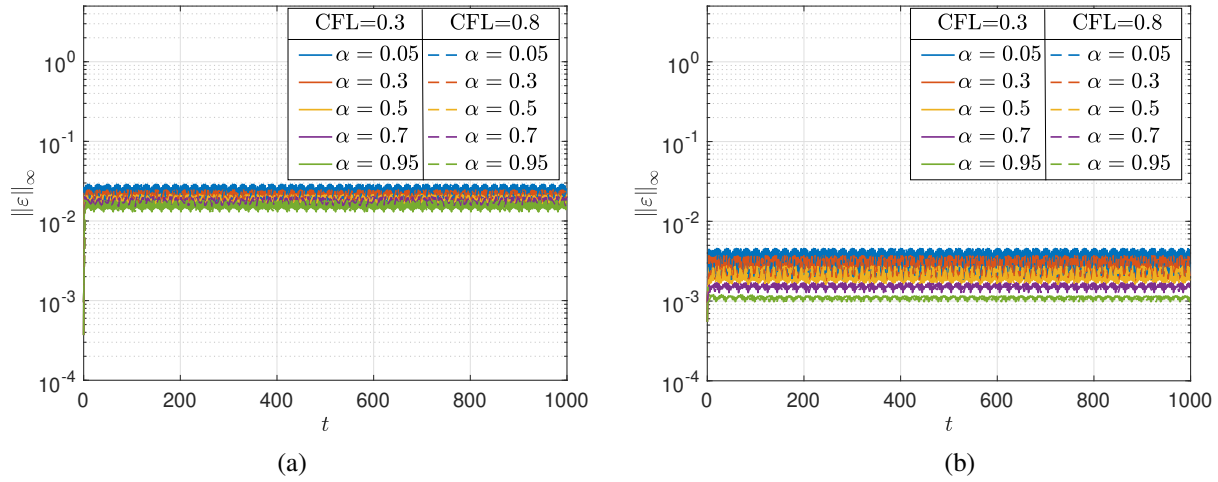


Fig. 3 L_∞ -error from a long-time simulation of (4) using $x_0 = (1 - \alpha)\Delta x = x_1 - \alpha\Delta x$, initial and boundary condition given by 59, various values of α , and (a) 1 - 2 - 1 and (b) 2 - 4 - 2 stencils for first-derivative approximation at two CFL numbers.

B. 2-D Euler equations: Isentropic convecting vortex

This section discusses the application of the cut-cell stencils of section (III.A.3) to solve the Euler equations. The conservative form of the two-dimensional Euler equations is given by

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0, \quad (60)$$

$$Q = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}, \quad F = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E + p) \end{bmatrix}, \quad G = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E + p) \end{bmatrix}, \quad (61)$$

$$E = \frac{p}{\gamma - 1} + \rho \left(\frac{u^2 + v^2}{2} \right), \quad (62)$$

where u, v are the Cartesian velocity components, ρ denotes the density, p is the pressure and E is the total energy. γ denotes the ratio of specific heats.

We solve the Euler equations for propagation of a compressible isentropic vortex on domains with different cut boundary shapes, as shown in figure 5. These cut boundaries help determine the sensitivity of the developed scheme to various cut-cell configurations. The domain boundaries in x -direction are assumed to be periodic, which allows the vortex to loop through the domain multiple times, assessing long-time stability of the method. Characteristic boundary conditions are applied in y -direction using the exact solution for strong BC enforcement.

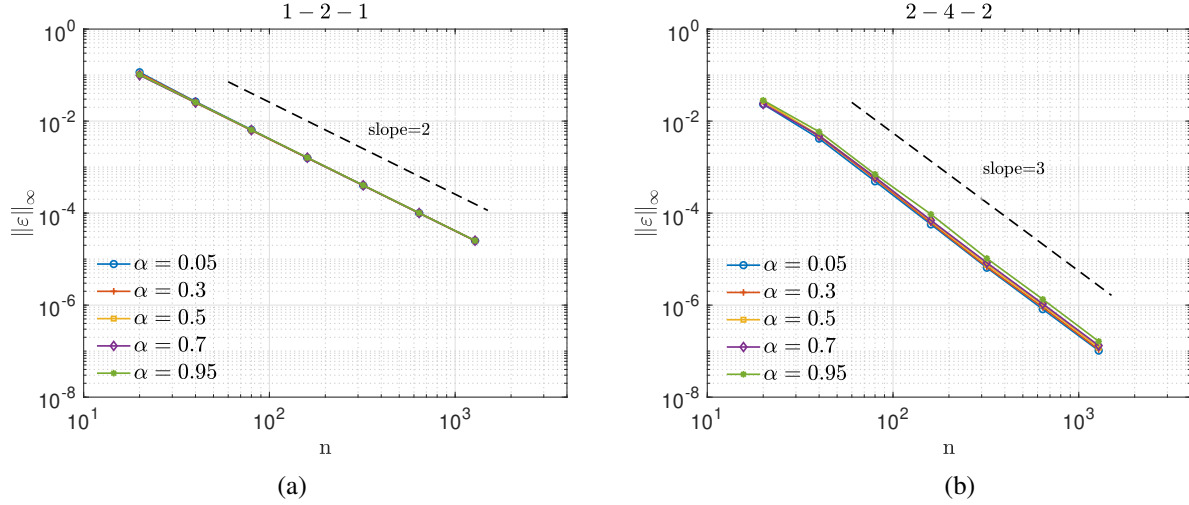


Fig. 4 L_∞ -norm of the solution error and convergence rate with the (a) 1 – 2 – 1 and (b) 2 – 4 – 2 stencils used to solve (4) with $x_0 = (1 - \alpha)\Delta x = x_1 - \alpha\Delta x$ and a domain with n grid points. Error calculations performed at $t_f = 1.0$. Dashed black line show the expected order-of-accuracy.

The exact solution to the problem is given by

$$\begin{aligned} \rho &= \left(1 - \frac{\varpi^2(\gamma - 1)}{8\pi^2 c_0^2} e^{1-\varphi^2 r^2}\right)^{\frac{1}{\gamma-1}}, & u &= u_0 - \frac{\varpi}{2\pi} \varphi(y - y_0 - v_0 t) e^{\frac{1-\varphi^2 r^2}{2}}, \\ v &= v_0 + \frac{\varpi}{2\pi} \varphi(x - x_0 - u_0 t) e^{\frac{1-\varphi^2 r^2}{2}}, & E &= \frac{p}{\gamma - 1} + \frac{1}{2} \rho(u^2 + v^2), \\ p &= \rho^\gamma, & r^2 &= (x - x_0 - u_0 t)^2 + (y - y_0 - v_0 t)^2, \end{aligned} \quad (63)$$

where (x_0, y_0) denotes the initial position of the vortex, (u_0, v_0) denotes the vortex convective velocity, φ is a scaling factor and ϖ denotes the non-dimensional circulation. We use $v_0 = 0$, $\gamma = 1.4$, $\varphi = 11$ and $\varpi = 1$. All quantities in (63) are non-dimensional, obtained from the density scale $= \rho_0^*$, velocity scale $u_0^* = \frac{c_0^*}{\sqrt{\gamma}}$, unit length scale and pressure scale $= \rho_0^* u_0^{*2}$, where $*$ denotes the dimensional quantities. The non-dimensional ambient speed of sound is $c_0 = \sqrt{\gamma}$.

The density and velocity magnitude error in figure 6 for various cut-cell configurations (depicted in figure 5) show only minor differences from one configuration to another, thus indicating that the developed method is not sensitive to cut-cell geometries. Figures 7(a) and (b) show the L_∞ -error from a long-time simulation using the 1 – 2 – 1 and the 2 – 4 – 2 scheme, respectively. A subsonic ($u_0 = 1.0$) and a supersonic ($u_0 = 2.0$) convective velocity is used to examine the robustness of boundary implementation for cases where all characteristics leave/enter the domain as well as cases where some characteristics enter, while the others leave the domain. The error remains constant with time indicating a time-stable behavior. As expected, errors from the 2 – 4 – 2 scheme are smaller than that from the 1 – 2 – 1 scheme. The calculations were performed at a CFL number of 0.5, showing for this non-linear problem that the method does not have a small-cell problem.

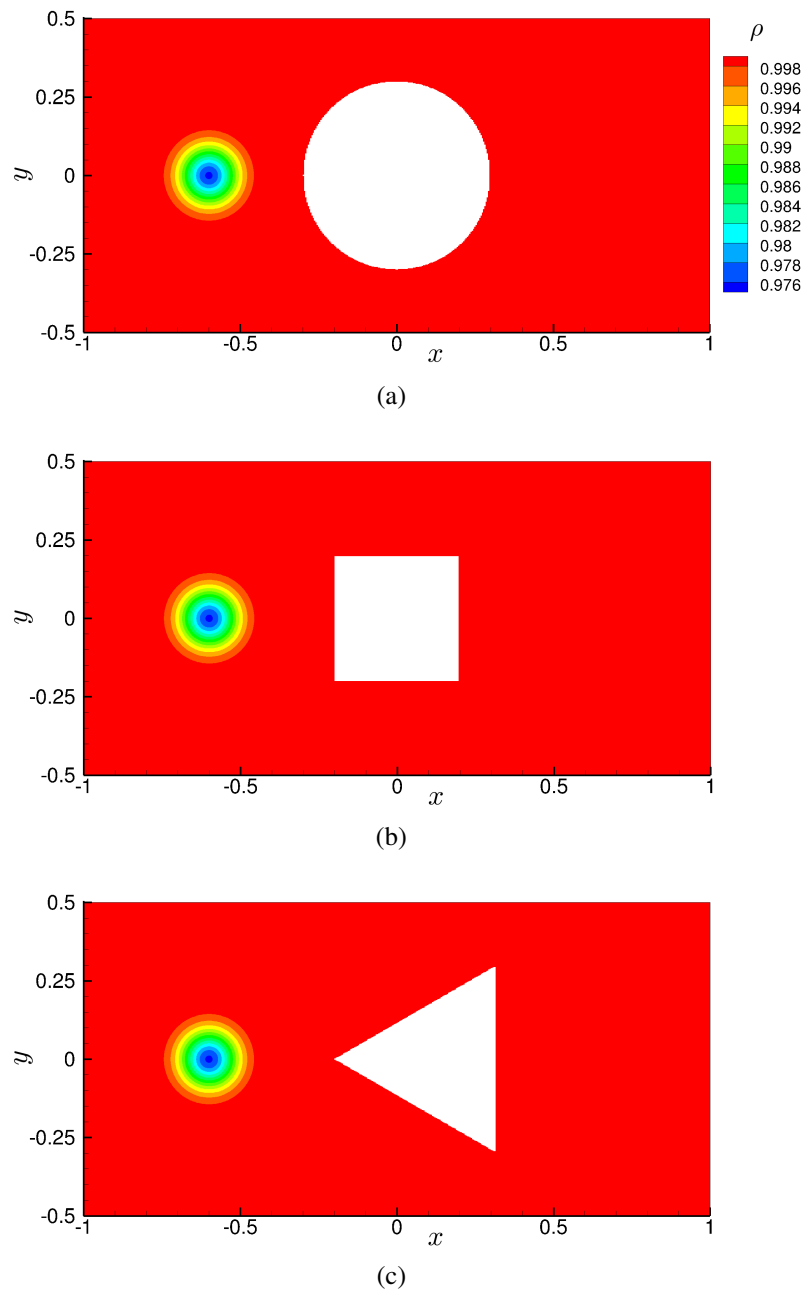


Fig. 5 Initial density contour for isentropic convecting vortex on various cut-cell domains: (a) Circle, (b) Square, and (c) Triangle. Legend is the same for all plots.

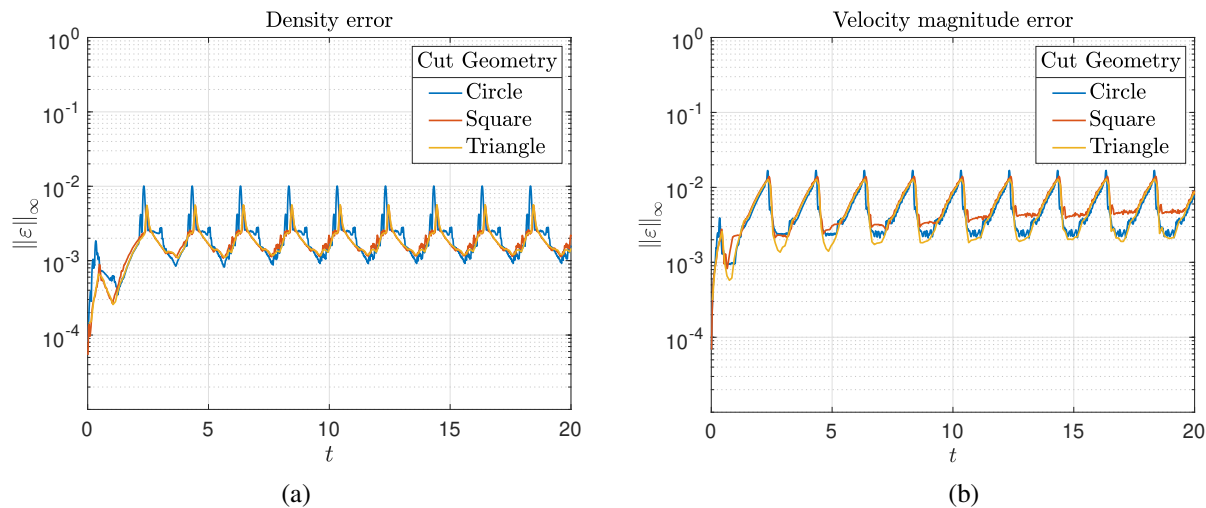


Fig. 6 L_∞ -error of (a) density and (b) velocity magnitude from solving the two-dimensional Euler equations (using the 2-4-2 stencil) on various cut-cell grids shown in figure 5 for the isentropic convecting vortex problem at subsonic convective velocity, $u_0 = 1$.

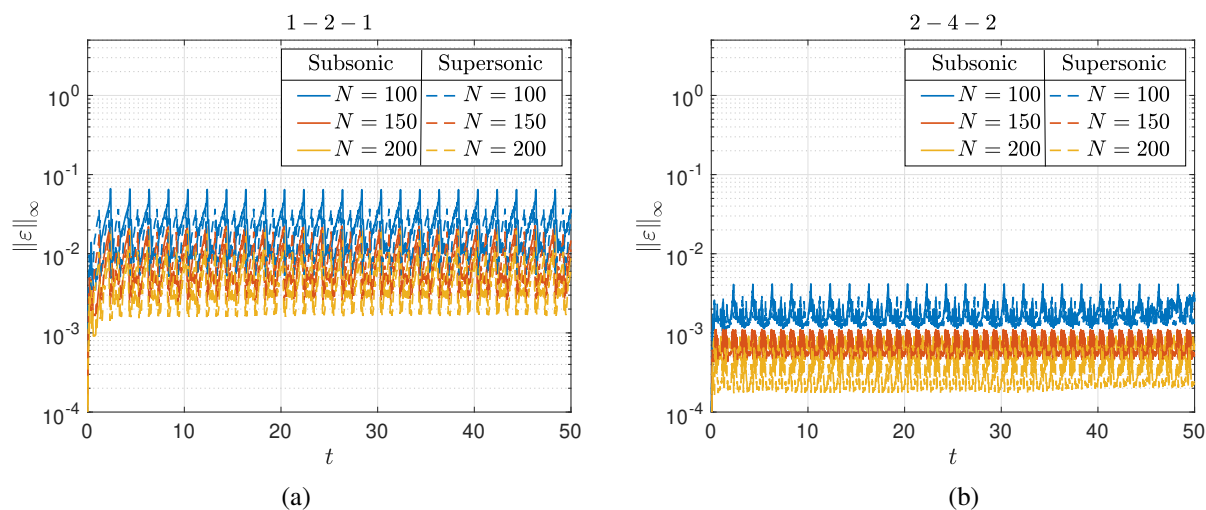


Fig. 7 L_∞ -error from a long-time simulation of the Euler equations for the convecting vortex problem on the cut-cell grid shown in figure 5(a) at subsonic and supersonic convective velocity using $N \times N$ grid points with (a) 1-2-1 and (b) 2-4-2 scheme.

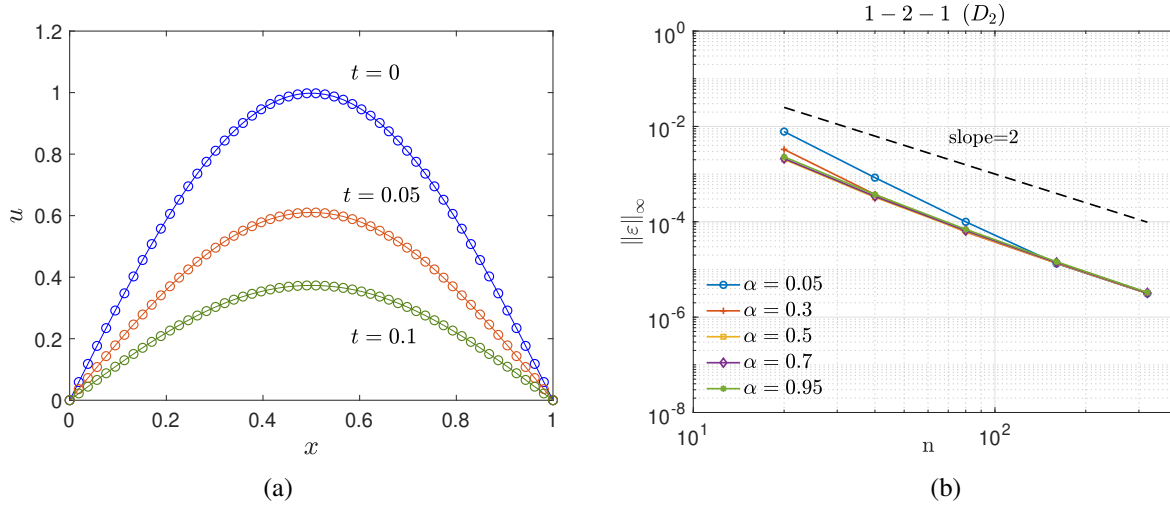


Fig. 8 (a) Solution of the heat equation (54) for the initial/boundary condition (64) and (b) L_∞ -norm of the solution error and convergence rate with the 1-2-1 (D_2) stencil.

C. 1-D heat equation

Consider the heat equation (54), on a spatial domain $x_0 \leq x \leq x_n = 1$, where $x_0 = (1 - \alpha)\Delta x = x_1 - \alpha\Delta x$. The initial and boundary conditions are

$$u(x, 0) = \sin \pi x, \quad u(x_0, t) = e^{-\pi^2 t} \sin \pi x_0, \quad u(1, t) = 0. \quad (64)$$

The exact solution to the problem is $u(x, t) = e^{-\pi^2 t} \sin \pi x$. Numerical solution at various times, using the 1-2-1 (D_2) stencil, is shown in figure 8(a). BC is strongly applied by injecting the boundary data. The L_∞ -error and the convergence rate of 1-2-1 (D_2) scheme for various values of α is depicted in figure 8(b). The scheme converges with second-order accuracy globally.

D. 2-D advection-diffusion equation

We solve the two-dimensional advection-diffusion equation on a cut-cell grid shown in figure 9(a). The governing equation is given by

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \mu \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right), \quad -L \leq x, y \leq L \quad t \geq 0, \quad (65)$$

$$u(x, y) = \frac{\partial r}{\partial x}, \quad v(x, y) = \frac{\partial r}{\partial y}, \quad (66)$$

$$r(x, y) = \sqrt{(x - x_0)^2 + (y - y_0)^2}, \quad (67)$$

where $L = 1.5$, $x_0 = 0$ and $y_0 = 0$. Dirichlet boundary condition is applied at the cut-cell boundary, $r = \sqrt{(x - x_0)^2 + (y - y_0)^2} = r_0$, given by

$$T(r = r_0, t) = \sin 2\pi (r_0 - t), \quad (68)$$

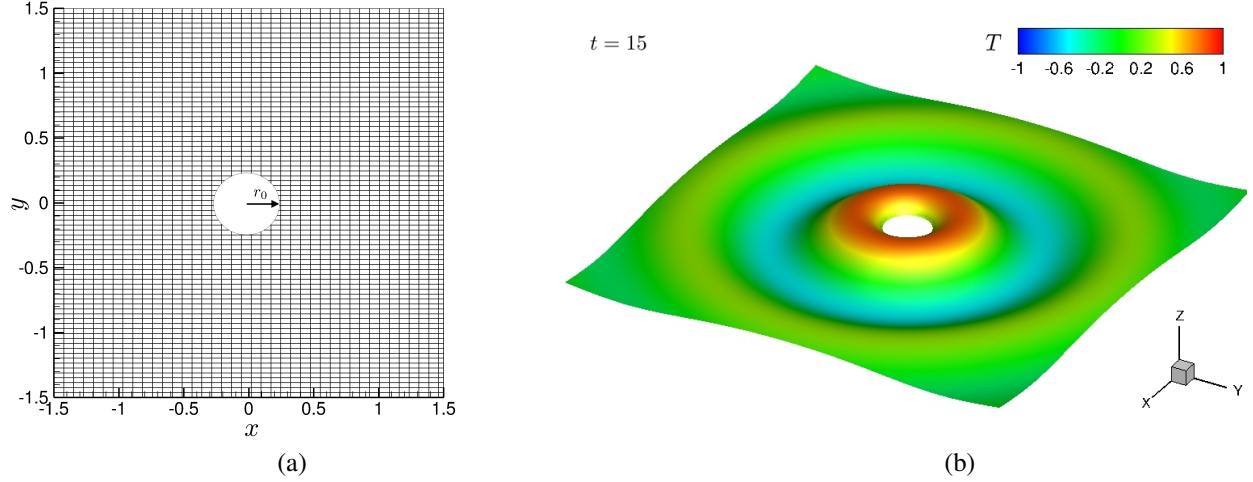


Fig. 9 (a) An example cut-cell grid for the advection-diffusion problem (66) and (b) a surface plot of the solution at $t = 15$ using $\mu = 0.025$ on the cut-cell domain shown in subfigure (a).

which introduces a time-periodic pulse that propagates radially outwards, and hence the cut-cell boundary acts as an inflow. Homogeneous Neumann condition is applied at the outer boundaries, $x, y = \pm 1.5$, that act as an outflow, given by

$$\nabla T \cdot \mathbf{n}|_{x,y=\pm 1.5} = 0. \quad (69)$$

$T(x, y, t = 0) = 0$ is used to initialize the computation. Figure 9(b) shows a surface plot of the solution with $\mu = 0.025$. The amplitude of the pulse diminishes as it propagates radially outwards. Figure 10(a) shows the solution, T , along the positive x -axis at $y = 0$ for various values of μ . As expected, increasing μ causes greater damping. Convergence of the $1 - 2 - 1$ and $1 - 2 - 1$ (D_2) stencils for this problem is examined in Figure 10(b), where the error, ε , is calculated at $t = 5$ against a very fine grid solution; the expected convergence rate is observed.

V. Conclusions

A systematic approach to derive high-order finite-difference cut-cell scheme is presented for efficient fluid-flow simulations over practical geometries. Constraints on cut-cell boundary stencils (with centered interior scheme) are derived to ensure long-time simulation stability without introducing ad hoc stabilization measures, *e.g.* artificial dissipation or numerical filter. Further, a simple solution to the small-cell problem is proposed by recognizing that the cell size in a conservative finite-difference scheme may not necessarily be equal to the grid spacing. First-derivative approximations that satisfy the summation-by-parts property for cut-cell grids are derived for second and fourth-order accuracy in interior and second and third-order accuracy globally. In addition, a second-derivative approximation of second-order accuracy in the interior and globally is presented. The derivation of high-order stencils will be a subject of future publication. The performance of the stencils presented here are evaluated for inviscid as well as viscous problems.

Acknowledgements

This work was supported by the US Department of Energy through the Los Alamos National Laboratory. Los Alamos National Laboratory is operated by Triad National Security, LLC, for the National Nuclear Security Administration of

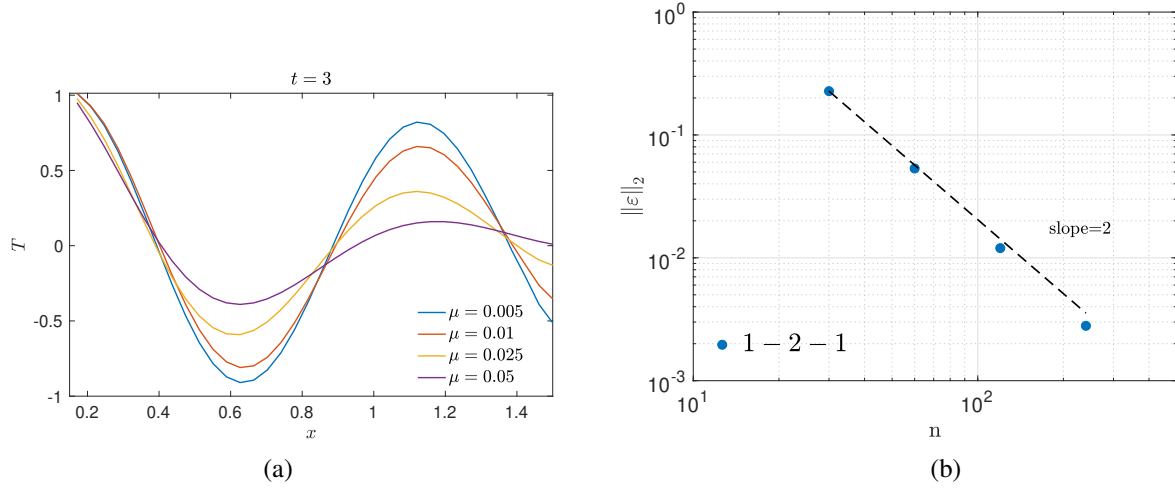


Fig. 10 (a) Solution of (66) along the positive x -axis at $y = 0$ for various values of μ with initial and boundary conditions as specified in section IV.D and (b) L_2 -norm of the solution error and convergence rate with the $1 - 2 - 1$ stencil for first-derivative approximation and $1 - 2 - 1$ (D_2) stencil for second-derivative approximation.

U.S. Department of Energy (Contract No. 89233218CNA000001). Research presented in this article was supported by the Laboratory Directed Research and Development program of Los Alamos National Laboratory under project number 20190227ER.

A. $2 - 4 - 2$ cut-cell stencil for first-derivative approximation

$$h_{00} = \gamma$$

$$h_{11} = \frac{1}{48} \left(2\alpha^4 - 8\alpha^3(\gamma - 2) + \alpha^2(44 - 48\gamma) + \alpha(48 - 88\gamma) - 48\gamma + 17 \right)$$

$$h_{22} = \frac{1}{48} \left(-6\alpha^4 + 8\alpha^3(3\gamma - 5) + 24\alpha^2(5\gamma - 3) + 144\alpha\gamma + 59 \right)$$

$$h_{33} = \frac{1}{48} \left(6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43 \right)$$

$$h_{44} = \frac{1}{48} \left(-2\alpha^4 + 8\alpha^3(\gamma - 1) + 8\alpha^2(3\gamma - 1) + 16\alpha\gamma + 49 \right)$$

$$d_{00} = -\frac{1}{2\gamma}$$

$$d_{01} = \frac{-2\alpha^4 k + 8\alpha^3(\gamma - 1)k + 4\alpha^2((6\gamma - 2)k + 3) + 4\alpha(4\gamma(k - 3) + 9) + 49k - 72\gamma + 24}{48\gamma}$$

$$d_{02} = -\frac{-2\alpha^4 k + 8\alpha^3(\gamma - 1)k + 8\alpha^2((3\gamma - 1)k + 1) + 16\alpha(\gamma(k - 2) + 1) + 49k - 32\gamma}{16\gamma}$$

$$d_{03} = \frac{-2\alpha^4 k + 8\alpha^3(\gamma - 1)k + 4\alpha^2((6\gamma - 2)k + 1) + 4\alpha(4\gamma(k - 1) + 1) + 49k - 8\gamma}{16\gamma}$$

$$d_{04} = \frac{k(2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49)}{48\gamma}$$

$$d_{05} = 0$$

$$d_{06} = 0$$

$$d_{10} = -\frac{-2\alpha^4 k + 8\alpha^3(\gamma - 1)k + 4\alpha^2((6\gamma - 2)k + 3) + 4\alpha(4\gamma(k - 3) + 9) + 49k - 72\gamma + 24}{2\alpha^4 - 8\alpha^3(\gamma - 2) + \alpha^2(44 - 48\gamma) + \alpha(48 - 88\gamma) - 48\gamma + 17}$$

$$d_{11} = 0$$

$$d_{12} = \frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 9)k + \alpha^4((64\gamma - 60)k + 2) + 8\alpha^3((23\gamma - 11)k - \gamma + 2) + \alpha^2((224\gamma + 1)k - 72\gamma + 56) + \alpha((96\gamma + 245)k - 208\gamma + 96) + 294k - 192\gamma + 59}{2(2\alpha^4 - 8\alpha^3(\gamma - 2) + \alpha^2(44 - 48\gamma) + \alpha(48 - 88\gamma) - 48\gamma + 17)}$$

$$d_{13} = -\frac{-2\alpha^6 k + 8\alpha^5(\gamma - 2)k + \alpha^4((56\gamma - 46)k + 4) + 4\alpha^3(2(17\gamma - 7)k - 4\gamma + 5) + \alpha^2((136\gamma + 25)k - 72\gamma + 28) + 4\alpha((12\gamma + 49)k - 20\gamma + 3) + 147k - 24\gamma + 4}{2\alpha^4 - 8\alpha^3(\gamma - 2) + \alpha^2(44 - 48\gamma) + \alpha(48 - 88\gamma) - 48\gamma + 17}$$

$$d_{14} = \frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 7)k + 6\alpha^4((8\gamma - 6)k + 1) + 8\alpha^3((13\gamma - 5)k - 3\gamma + 3) + 3\alpha^2((32\gamma + 11)k - 24\gamma + 8) + \alpha((32\gamma + 147)k - 48\gamma) + 98k - 3}{2(2\alpha^4 - 8\alpha^3(\gamma - 2) + \alpha^2(44 - 48\gamma) + \alpha(48 - 88\gamma) - 48\gamma + 17)}$$

$$d_{15} = 0$$

$$d_{16} = 0$$

$$d_{20} = -\frac{3(2\alpha^4 k - 8\alpha^3(\gamma - 1)k - 8\alpha^2((3\gamma - 1)k + 1) - 16\alpha(\gamma(k - 2) + 1) - 49k + 32\gamma)}{-6\alpha^4 + 8\alpha^3(3\gamma - 5) + 24\alpha^2(5\gamma - 3) + 144\alpha\gamma + 59}$$

$$d_{21} = -\frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 9)k + \alpha^4((64\gamma - 60)k + 2) + 8\alpha^3((23\gamma - 11)k - \gamma + 2) + \alpha^2((224\gamma + 1)k - 72\gamma + 56) + \alpha((96\gamma + 245)k - 208\gamma + 96) + 294k - 192\gamma + 59}{2(-6\alpha^4 + 8\alpha^3(3\gamma - 5) + 24\alpha^2(5\gamma - 3) + 144\alpha\gamma + 59)}$$

$$d_{22} = 0$$

$$d_{23} = -\frac{6\alpha^6 k - 6\alpha^5(4\gamma - 7)k - 6\alpha^4(8(3\gamma - 2)k + 3) - 8\alpha^3((33\gamma - 9)k - 9\gamma + 10) - 3\alpha^2((48\gamma + 49)k - 88\gamma + 24) - 9\alpha(49k - 16\gamma) - 59}{2(-6\alpha^4 + 8\alpha^3(3\gamma - 5) + 24\alpha^2(5\gamma - 3) + 144\alpha\gamma + 59)}$$

$$d_{24} = \frac{\alpha(\alpha + 2)(2\alpha^4 k - 8\alpha^3(\gamma - 1)k - 8\alpha^2((3\gamma - 1)k + 1) - 16\alpha(\gamma(k - 2) + 1) - 49k + 32\gamma)}{-6\alpha^4 + 8\alpha^3(3\gamma - 5) + 24\alpha^2(5\gamma - 3) + 144\alpha\gamma + 59}$$

$$d_{25} = 0$$

$$d_{26} = 0$$

$$d_{30} = \frac{3(2\alpha^4 k - 8\alpha^3(\gamma - 1)k - 4\alpha^2((6\gamma - 2)k + 1) - 4\alpha(4\gamma(k - 1) + 1) - 49k + 8\gamma)}{6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43}$$

$$d_{31} = \frac{-2\alpha^6 k + 8\alpha^5(\gamma - 2)k + \alpha^4((56\gamma - 46)k + 4) + 4\alpha^3(2(17\gamma - 7)k - 4\gamma + 5) + \alpha^2((136\gamma + 25)k - 72\gamma + 28) + 4\alpha((12\gamma + 49)k - 20\gamma + 3) + 147k - 24\gamma + 4}{6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43}$$

$$d_{32} = -\frac{-6\alpha^6 k + 6\alpha^5(4\gamma - 7)k + 6\alpha^4(8(3\gamma - 2)k + 3) + 8\alpha^3((33\gamma - 9)k - 9\gamma + 10) + 3\alpha^2((48\gamma + 49)k - 88\gamma + 24) + 9\alpha(49k - 16\gamma) + 59}{2(6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43)}$$

$$d_{33} = 0$$

$$d_{34} = \frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 5)k + 2\alpha^4(8(2\gamma - 1)k + 5) + 8\alpha^3((5\gamma - 1)k - 5\gamma + 5) + \alpha^2((16\gamma + 49)k - 120\gamma + 40) + \alpha(49k - 80\gamma) + 59}{2(6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43)}$$

$$d_{35} = -\frac{4}{6\alpha^4 - 8\alpha^3(3\gamma - 4) + \alpha^2(36 - 96\gamma) - 72\alpha\gamma + 43}$$

$$d_{36} = 0$$

$$d_{40} = k$$

$$d_{41} = \frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 7)k + 6\alpha^4((8\gamma - 6)k + 1) + 8\alpha^3((13\gamma - 5)k - 3\gamma + 3) + 3\alpha^2((32\gamma + 11)k - 24\gamma + 8) + \alpha((32\gamma + 147)k - 48\gamma) + 98k - 3}{2(2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49)}$$

$$d_{42} = \frac{\alpha(\alpha + 2)(2\alpha^4 k - 8\alpha^3(\gamma - 1)k - 8\alpha^2((3\gamma - 1)k + 1) - 16\alpha(\gamma(k - 2) + 1) - 49k + 32\gamma)}{2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49}$$

$$d_{43} = \frac{-2\alpha^6 k + 2\alpha^5(4\gamma - 5)k + 2\alpha^4(8(2\gamma - 1)k + 5) + 8\alpha^3((5\gamma - 1)k - 5\gamma + 5) + \alpha^2((16\gamma + 49)k - 120\gamma + 40) + \alpha(49k - 80\gamma) + 59}{2(2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49)}$$

$$d_{44} = 0$$

$$d_{45} = -\frac{32}{2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49}$$

$$d_{46} = \frac{4}{2\alpha^4 - 8\alpha^3(\gamma - 1) + \alpha^2(8 - 24\gamma) - 16\alpha\gamma - 49}$$

For a copy of these coefficients in .txt or another format, please email Nek Sharan (nsharan@lanl.gov).

References

- [1] Ingram, D. M., Causon, D. M., and Mingham, C. G., “Developments in Cartesian cut cell methods,” *Mathematics and Computers in Simulation*, Vol. 61, No. 3-6, 2003, pp. 561–572.
- [2] Berger, M., and Aftosmis, M., “Progress towards a Cartesian cut-cell method for viscous compressible flow,” *50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition*, 2012, p. 1301.
- [3] Tseng, Y.-H., and Ferziger, J. H., “A ghost-cell immersed boundary method for flow in complex geometry,” *Journal of Computational Physics*, Vol. 192, No. 2, 2003, pp. 593–623.
- [4] Ghias, R., Mittal, R., and Dong, H., “A sharp interface immersed boundary method for compressible viscous flows,” *Journal of Computational Physics*, Vol. 225, No. 1, 2007, pp. 528–553.
- [5] Berger, M., and LeVeque, R., “Stable boundary conditions for Cartesian grid calculations,” *Computing Systems in Engineering*, Vol. 1, No. 2-4, 1990, pp. 305–311.
- [6] Hu, X. Y., Khoo, B., Adams, N. A., and Huang, F., “A conservative interface method for compressible flows,” *Journal of Computational Physics*, Vol. 219, No. 2, 2006, pp. 553–578.
- [7] Chung, M.-H., “Cartesian cut cell approach for simulating incompressible flows with rigid bodies of arbitrary shape,” *Computers & Fluids*, Vol. 35, No. 6, 2006, pp. 607–623.
- [8] Hartmann, D., Meinke, M., and Schröder, W., “A strictly conservative Cartesian cut-cell method for compressible viscous flows on adaptive grids,” *Computer Methods in Applied Mechanics and Engineering*, Vol. 200, No. 9-12, 2011, pp. 1038–1052.

- [9] Colella, P., Graves, D. T., Keen, B. J., and Modiano, D., "A Cartesian grid embedded boundary method for hyperbolic conservation laws," *Journal of Computational Physics*, Vol. 211, No. 1, 2006, pp. 347–366.
- [10] Yang, G., Causon, D., and Ingram, D., "Calculation of compressible flows about complex moving geometries using a three-dimensional Cartesian cut cell method," *International Journal for Numerical Methods in Fluids*, Vol. 33, No. 8, 2000, pp. 1121–1151.
- [11] Bakhtian, N. M., and Aftosmis, M. J., "Parametric Study of Peripheral Nozzle Configurations for Supersonic Retropropulsion," *Journal of Spacecraft and Rockets*, Vol. 47, No. 6, 2010, pp. 935–950.
- [12] Tan, S., Wang, C., Shu, C.-W., and Ning, J., "Efficient implementation of high order inverse Lax–Wendroff boundary treatment for conservation laws," *Journal of Computational Physics*, Vol. 231, No. 6, 2012, pp. 2510–2527.
- [13] Muralidharan, B., and Menon, S., "A high-order adaptive Cartesian cut-cell method for simulation of compressible viscous flow over immersed bodies," *Journal of Computational Physics*, Vol. 321, 2016, pp. 342–368.
- [14] Rai, M. M., and Moin, P., "Direct simulations of turbulent flow using finite-difference schemes," *Journal of Computational Physics*, Vol. 96, No. 1, 1991, pp. 15–53.
- [15] Morinishi, Y., Lund, T. S., Vasilyev, O. V., and Moin, P., "Fully conservative higher order finite difference schemes for incompressible flow," *Journal of Computational Physics*, Vol. 143, No. 1, 1998, pp. 90–124.
- [16] Ryu, J., and Livescu, D., "Turbulence structure behind the shock in canonical shock–vortical turbulence interaction," *Journal of Fluid Mechanics*, Vol. 756, 2014.
- [17] Sharan, N., Matheou, G., and Dimotakis, P. E., "Turbulent shear-layer mixing: initial conditions, and direct-numerical and large-eddy simulations," *Journal of Fluid Mechanics*, Vol. 877, 2019, pp. 35–81.
- [18] Tam, C. K., and Dong, Z., "Wall boundary conditions for high-order finite-difference schemes in computational aeroacoustics," *Theoretical and Computational Fluid Dynamics*, Vol. 6, No. 5-6, 1994, pp. 303–322.
- [19] Zhuang, M., and Chen, R., "Optimized upwind dispersion-relation-preserving finite difference scheme for computational aeroacoustics," *AIAA Journal*, Vol. 36, No. 11, 1998, pp. 2146–2148.
- [20] Colonius, T., and Lele, S. K., "Computational aeroacoustics: progress on nonlinear problems of sound generation," *Progress in Aerospace Sciences*, Vol. 40, No. 6, 2004, pp. 345–416.
- [21] Saurabh, S., "Direct numerical simulation of human phonation," Ph.D. thesis, University of Illinois at Urbana-Champaign, 2017.
- [22] Brady, P., and Livescu, D., "Foundations for high-order, conservative cut-cell methods: Stable discretizations on degenerate meshes," *Journal of Computational Physics*, 2020, p. 109794.
- [23] Brady, P. T., and Livescu, D., "High-order, stable, and conservative boundary schemes for central and compact finite differences," *Computers & Fluids*, Vol. 183, 2019, pp. 84–101.
- [24] Sharan, N., Brady, P., and Livescu, D., "High-order energy-stable boundary treatment for finite-difference cut-cell method," *Bulletin of the American Physical Society*, 2019.
- [25] Sharan, N., Brady, P. T., and Livescu, D., "Stable and conservative boundary treatment for difference methods, with application to cut-cell discretizations," *AIAA Scitech 2020 Forum*, 2020, p. 0807.
- [26] Fernández, D. C. D. R., Hicken, J. E., and Zingg, D. W., "Review of summation-by-parts operators with simultaneous approximation terms for the numerical solution of partial differential equations," *Computers & Fluids*, Vol. 95, 2014, pp. 171–196.

- [27] Sharan, N., “Time-stable high-order finite difference methods for overset grids,” Ph.D. thesis, University of Illinois at Urbana-Champaign, 2016.
- [28] Sharan, N., Pantano, C., and Bodony, D. J., “Time-stable overset grid method for hyperbolic problems using summation-by-parts operators,” *Journal of Computational Physics*, Vol. 361, 2018, pp. 199–230.
- [29] Schneiders, L., Hartmann, D., Meinke, M., and Schröder, W., “An accurate moving boundary formulation in cut-cell methods,” *Journal of Computational Physics*, Vol. 235, 2013, pp. 786–809.
- [30] Mohammadi-Aragh, M., Klingbeil, K., Brüggemann, N., Eden, C., and Burchard, H., “The impact of advection schemes on restratification due to lateral shear and baroclinic instabilities,” *Ocean Modelling*, Vol. 94, 2015, pp. 112–127.
- [31] Sharan, N., Matheou, G., and Dimotakis, P., “Effects of numerical dissipation and unphysical excursions on scalar-mixing estimates in large-eddy simulations,” *APS*, 2017, pp. L31–007.
- [32] Sharan, N., Matheou, G., and Dimotakis, P. E., “Mixing, scalar boundedness, and numerical dissipation in large-eddy simulations,” *Journal of Computational Physics*, Vol. 369, 2018, pp. 148–172.
- [33] Reddy, S. C., and Trefethen, L. N., “Stability of the method of lines,” *Numerische Mathematik*, Vol. 62, No. 1, 1992, pp. 235–267.
- [34] Kreiss, H.-O., and Scherer, G., “Method of lines for hyperbolic differential equations,” *SIAM Journal on Numerical Analysis*, Vol. 29, No. 3, 1992, pp. 640–646.
- [35] Gustafsson, B., *High order difference methods for time dependent PDE*, Vol. 38, Springer Science & Business Media, 2007.
- [36] Antsaklis, P. J., and Michel, A. N., *A Linear Systems Primer*, 1st ed., Birkhäuser Basel, 2007.
- [37] Corless, M. J., and Frazho, A., *Linear systems and control: an operator perspective*, CRC Press, 2003.
- [38] Sharan, N., Pantano, C., and Bodony, D. J., “Energy stable overset grid methods for hyperbolic problems,” *7th AIAA Theoretical Fluid Mechanics Conference*, 2014, p. 2924.
- [39] Kreiss, H.-O., and Scherer, G., “Finite element and finite difference methods for hyperbolic partial differential equations,” *Mathematical Aspects of Finite Elements in Partial Differential Equations*, Academic Press, 1974, pp. 195–212.
- [40] Carpenter, M. H., Gottlieb, D., and Abarbanel, S., “Time-stable boundary conditions for finite-difference schemes solving hyperbolic systems: methodology and application to high-order compact schemes,” *Journal of Computational Physics*, Vol. 111, No. 2, 1994, pp. 220–236.
- [41] Gustafsson, B., “The convergence rate for difference approximations to mixed initial boundary value problems,” *Mathematics of Computation*, Vol. 29, No. 130, 1975, pp. 396–406.